

Name	Quantity	Unit
electric field intensity	$\vec{E}$	V/m
electric flux density	$\vec{D}$	C/m ²
magnetic field intensity	$\vec{H}$	A/m
magnetic flux density	$\vec{B}$	T (teslas)
permeability of free space	μ_0	H/m
permittivity of free space	ϵ_0	F/m
dielectric permittivity	ϵ	F/m
electric flux	ϕ	V·m
electric potential	V_A	V
electric dipole	$\vec{p}$	C·m
capacitance	C	F
electrostatic energy density	w_e	
conductive current density	$\vec{J}$	A/m ²
electric conductivity	σ	($\Omega \cdot m$) ⁻¹
electric resistivity	ρ	$\Omega \cdot m$
magnetic flux	ψ	Wb (webers)
wavelength	λ	m
speed of light	c	m/s
frequency	f	Hz
angular frequency	ω	rad/s
resistance	R	Ω
self-inductance	L	H
capacitance	C	F
active power	P	W
reactive power	Q	VAR

apparent power	S	VA
space angle	δ	rad
phase shift	φ	rad
impedance	Z	Ω
reactance	X	Ω
conductance	G	$S(\text{siemens}) = \Omega^{-1}$
susceptance	B	$\mathcal{S}$
admittance	Y	S
power factor	k	—
magnetic coupling coeff.	k	—
mutual coupling	M	H
damping/alternation constant	α	—
phase constant	β	
distortion factor	k_d	
peak factor	k_p	
deformant power	D	VA def
electromotive force	e	V
magnetization	$\overline{H}$	A/m
charge density, volume	ρ_v	C/m ³

Curs 1

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} ; \vec{B} = \mu_0 \cdot \vec{H} ; \vec{D} = \epsilon_0 \cdot \vec{E}$$

$$i = - \frac{dQ}{dt}$$

$$Q = \int_S \vec{g}_s \cdot d\vec{A}$$

$$Q = \int_c \vec{g}_e \cdot d\vec{s}$$

$$Q = \int_v \vec{g}_v \cdot d\vec{v}$$

$$\vec{E} = \vec{E}_n \cdot \vec{E}_o$$

Curs 2

$$\Phi = \iint_S \vec{E} \cdot d\vec{A}$$

$$\Phi_E = \frac{Q}{\epsilon_0}$$

$$\vec{p} = Q \cdot \vec{r}$$

Curs 3

$$C = \frac{Q}{U}$$

$$w_e = \frac{1}{2} \cdot \vec{D} \cdot \vec{E}$$

$$\vec{J} = \nabla \cdot \vec{E}$$

$$\vec{E} = \vec{g} \cdot \vec{J}$$

$$\vec{g} = \frac{1}{\nabla}$$

Curs 4

$$\Psi = \iint_S \vec{B} \cdot d\vec{A}$$

$$\text{rot } \vec{H} = \vec{J}$$

$$\text{rot } \vec{E} = 0$$

Curs 5

$$\omega = 2\pi f$$

$$\lambda = \frac{c}{f}$$

$$\text{Ohm: } R \cdot i = u_B$$

$$P = U_B \cdot i = R \cdot i^2$$

$$u(t) = U\sqrt{2} \sin(\omega t + \gamma)$$

$$i(t) = I\sqrt{2} \sin(\omega t + \delta)$$

$$Z = \sqrt{R^2 + X^2}$$

$$X = \omega L - \frac{1}{\omega C} = X_L - X_C$$

$$G = \frac{1}{R}$$

$$\frac{1}{\omega L} - \omega C = B_L - B_C = B$$

$$Y = \sqrt{G^2 + B^2}$$

$$P = U \cdot I \cdot \cos \varphi$$

$$Q = U \cdot I \cdot \sin \varphi$$

$$S = \sqrt{P^2 + Q^2} = U \cdot I$$

$$k = \frac{P}{S}$$

$$\underline{I} = I \cdot e^{j\delta}$$

Ques 7

$$R = \frac{H}{\sqrt{L_1 L_2}}$$

$$\underline{AD} - \underline{BC} = 1 \quad \text{-reciprocity condition}$$

Ques 8

$$d = \ln \frac{U_1}{U_2}$$

$$\beta = j \cdot \gamma$$

$$R_d = \frac{\sqrt{I_2^2 + I_3^2 + \dots}}{\sqrt{I_1^2 + I_2^2 + \dots}}$$

$$R_p = \frac{I_{\max}}{I_{RMS}}$$

$$D = \sqrt{S^2 - P^2 - Q^2}$$

$$R_p = \frac{P}{S} = \cos \varphi$$